Kaili Cao

Curriculum Vitae

Michigan Institute for Data and AI in Society
University of Michigan, Ann Arbor
✉ *kailicao@umich.edu*
kailicao.github.io
Github *LinkedIn*

Education

2023 – 2026 **Doctor of Philosophy, Physics**, *The Ohio State University*, Columbus, United States.
Thesis: *Renovating Analysis Infrastructure for the Next Generation of Astronomical Surveys*

2021 – 2023 **Master of Science, Physics**, *The Ohio State University*, Columbus, United States.

2020 – 2021 **31 credits, Computer Science and Technology**, *Nanjing University*, Nanjing, China.

2016 – 2020 **Bachelor of Arts, French Language and Literature**, *Nanjing University*, Nanjing, China.

Positions

2026 – 2028 **Schmidt AI in Science Postdoctoral Fellow**, *University of Michigan*, Ann Arbor, United States.

2024 – present **Member**, *Asteroseismology Using The Galactic Bulge Time Domain Survey (Roman Space Telescope Wide-Field Science Team)*, Columbus, United States.

2023 – present **Member**, *Maximizing Cosmological Science with the Roman High Latitude Imaging Survey (Roman Space Telescope Project Infrastructure Team)*, Pasadena, United States.

2019 **Visiting Undergraduate Student**, *Massachusetts Institute of Technology*, Cambridge, United States.

Interests and Skills

Cosmology	Weak Gravitational Lensing, Cosmic Shear Measurement, Parameter Inference, Fisher Forecasts
Stellar Physics	Stellar Models, Stellar Evolution Code, Interior Physics, Asteroseismology, Red Giant Branch
Methods	Numerical Simulations, Numerical Methods, Image Processing, Data Analysis, Machine Learning
Programming	Python (main), C/C++, Fortran, Shell/Perl, Wolfram Mathematica
Languages	Chinese (native), English (working proficiency), French (undergraduate major)
Certificates	Productivity Program (The Ohio State University, 2021), Responsible Conduct of Research (The Ohio State University, 2024), Data Science Boot Camp (The Erdős Institute, 2025)

Publications

ORCID ORCID ID: 0000-0002-1699-6944; NASA/ADS: orcid:0000-0002-1699-6944

First-Author Publications

Jul 2026 **Kaili Cao**, Efficient Optimal Image Reconstruction for the Nancy Grace Roman Space Telescope and Beyond: II. Implementation of Effortless, *arXiv e-prints*, arXiv:2607.09862 [astro-ph.IM], July 2026, doi:10.48550/arXiv.2607.09862 [GitHub]

Jul 2026 **Kaili Cao**, Efficient Optimal Image Reconstruction for the Nancy Grace Roman Space Telescope and Beyond: I. First Results with Effortless, *arXiv e-prints*, arXiv:2607.06674 [astro-ph.IM], July 2026, doi:10.48550/arXiv.2607.06674 [GitHub]

Mar 2026 **Kaili Cao** and Roman HLIS Cosmology PIT, Linear Image Regridding and Coaddition with Oversampled Point Spread Functions: Lessons from 1D, *The Astronomical Journal*, Volume 171, Number 3, March 2026, Page 140, doi:10.3847/1538-3881/ae356a [GitHub]

Feb 2026 **Kaili Cao**, Christopher M. Hirata, Katherine Laliotis, et al., Simulating Image Coaddition with the Nancy Grace Roman Space Telescope. IV. Hyperparameter Optimization and Experimental Features, *The Astrophysical Journal*, Volume 998, Number 2, February 2026, Page 304, doi:10.3847/1538-4357/ae3a97 [GitHub]

Jan 2026 **Kaili Cao** and Marc H. Pinsonneault, Modeling APOKASC-3 Red Giants. I. The First Dredge-up and Red Giant Branch Bump, *The Astrophysical Journal*, Volume 997, Number 1, January 2026, Page 120, doi:10.3847/1538-4357/ae22d1 [GitHub, Zenodo]

Jan 2026 (accepted) **Kaili Cao**, David H. Weinberg, Vivian Miranda, et al., Fisher Forecasts for Cosmological Yields from 3×2pt Analysis of the Roman Space Telescope High Latitude Imaging Survey, *arXiv e-prints*, arXiv:2601.00438 [astro-ph.CO], January 2026, doi:10.48550/arXiv.2601.00438 [accepted for publication]

Dec 2025 **Kaili Cao**, ContEvol Formalism: Numerical Methods Based on Hermite Spline Optimization, *Mathematics*, Volume 13, Issue 24, December 2025, Article Number 3981, doi:10.3390/math13243981 [GitHub]

Sep 2025 **Kaili Cao** and Marc H. Pinsonneault, Toward Accurate Asteroseismic Masses for Luminous Giants, *The Astrophysical Journal Letters*, Volume 990, Number 1, September 2025, Page L26, doi:10.3847/2041-8213/adfa12

Apr 2025 **Kaili Cao**, Christopher M. Hirata, Katherine Laliotis, et al., Simulating Image Coaddition with the Nancy Grace Roman Space Telescope. III. Software Improvements and New Linear Algebra Strategies, *The Astrophysical Journal Supplement Series*, Volume 277, Number 2, April 2025, Page 55, doi:10.3847/1538-4365/adb580 [GitHub]

May 2021 **Kaili Cao**, David J. Barnes, and Mark Vogelsberger, Studying galaxy cluster morphological metrics with MOCK-X, *Monthly Notices of the Royal Astronomical Society*, Volume 503, Issue 3, May 2021, Pages 3394–3413, doi:10.1093/mnras/stab605

Co-authored Publications in Cosmology

Jul 2026 Chun-Hao To, Tae-Hyeon Shin, Chihway Chang, et al. (with **Kaili Cao** as 10th of 21 authors), Robust Photometry for Roman High-Latitude Imaging Survey Cosmology Using Roman and Rubin Imaging, *arXiv e-prints*, arXiv:2607.09849 [astro-ph.IM], July 2026, doi:10.48550/arXiv.2607.09849

Jun 2026 Junzhou Zhang, Chihway Chang, Jiachuan Xu, et al. (with **Kaili Cao** as 7th of 9 authors), Accurate modeling for 3 × 2pt analyses in Roman and Rubin: a study of model approximations, *arXiv e-prints*, arXiv:2606.23781 [astro-ph.CO], June 2026, doi:10.48550/arXiv.2606.23781

Apr 2026 Katherine Laliotis, Christopher M. Hirata, Emily Macbeth, and **Kaili Cao**, Removing correlated noise stripes from the Nancy Grace Roman Space Telescope survey images, *Publications of the Astronomical Society of Japan*, Volume 78, Issue 2, April 2026, Page 810–823, doi:10.1093/pasj/psag020 [GitHub]

Dec 2025 Openuniverse, Lsst Dark Energy Science Collaboration, Roman Hlis Project Infrastructure, et al. (authors in alphabetical order, including **K. Cao**), OpenUniverse2024: a shared, simulated view of the sky for the next generation of cosmological surveys, *Monthly Notices of the Royal Astronomical Society*, Volume 544, Issue 4, December 2025, Pages 3799–3823, doi:10.1093/mnras/staf1833

Dec 2024 Katherine Laliotis, Emily Macbeth, Christopher M. Hirata, et al. (with **Kaili Cao** as 4th of 6 authors), Analysis of Biasing from Noise from the Nancy Grace Roman Space Telescope: Implications for Weak Lensing, *Publications of the Astronomical Society of the Pacific*, Volume 136, Number 12, December 2024, Page 124506, doi:10.1088/1538-3873/ad9bec

Mar 2024 Masaya Yamamoto, Katherine Laliotis, Emily Macbeth, et al. (with **Kaili Cao** as 7th of 19 authors), Simulating image coaddition with the Nancy Grace Roman Space Telescope - II. Analysis of the simulated images and implications for weak lensing, *Monthly Notices of the Royal Astronomical Society*, Volume 528, Issue 4, March 2024, Pages 6680–6705, doi:10.1093/mnras/stae177

Feb 2024 Christopher M. Hirata, Masaya Yamamoto, Katherine Laliotis, et al. (with **Kaili Cao** as 7th of 19 authors), Simulating image coaddition with the Nancy Grace Roman Space Telescope - I. Simulation methodology and general results, *Monthly Notices of the Royal Astronomical Society*, Volume 528, Issue 2, February 2024, Pages 2533–2561, doi:10.1093/mnras/stae182

Sep 2021 David J. Barnes, Mark Vogelsberger, Francesca A. Pearce, et al. (with **Kaili Cao** as 6th of 8 authors), Characterizing hydrostatic mass bias with MOCK-X, *Monthly Notices of the Royal Astronomical Society*, Volume 506, Issue 2, September 2021, Pages 2533–2550, doi:10.1093/mnras/stab1276

Co-authored Publications in Stellar Astrophysics

Jul 2026 Marc H. Pinsonneault, Jennifer L. van Saders, Lyra Cao, et al. (with **Kaili Cao** as 13th of 14 authors), The YREC Stellar Evolution Code: Public Data Release, *The Astrophysical Journal*, Volume 1006, Number 2, July 2026, Page 220, doi:10.3847/1538-4357/ae71cf [GitHub]

Oct 2025 B. Liagre, R. A. García, S. Mathur, et al. (with **K. Cao** as 7th of 12 authors), Beyond the Nyquist frequency: Asteroseismic catalog of undersampled Kepler late subgiants and early red giants, *Astronomy & Astrophysics*, Volume 702, October 2025, Article Number A144, doi:10.1051/0004-6361/202555167

Jul 2025 Trevor J. Weiss, Noah J. Downing, Marc H. Pinsonneault, et al. (with **Kaili Cao** as 7th of 13 authors), Modeling Asteroseismic Yields for the Roman Galactic Bulge Time-domain Survey, *The Astrophysical Journal*, Volume 987, Number 2, July 2025, Page 181, doi:10.3847/1538-4357/adde5b

May 2025 M. Vrard, M. H. Pinsonneault, Y. Elsworth, et al. (with **K. Cao** as 11th of 15 authors), Red giant evolutionary status determination: The complete Kepler catalog, *Astronomy & Astrophysics*, Volume 697, May 2025, Article Number A165, doi:10.1051/0004-6361/202452635

Feb 2025 Marc H. Pinsonneault, Joel C. Zinn, Jamie Tayar, et al. (with **Kaili Cao** as 19th of 41 authors), APOKASC-3: The Third Joint Spectroscopic and Asteroseismic Catalog for Evolved Stars in the Kepler Fields, *The Astrophysical Journal Supplement Series*, Volume 276, Number 2, February 2025, Page 69, doi:10.3847/1538-4365/ad9fef

May 2024 John D. Roberts, Marc H. Pinsonneault, Jennifer A. Johnson, et al. (with **Kaili Cao** as 11th of 19 authors), Nature versus nurture: distinguishing effects from stellar processing and chemical evolution on carbon and nitrogen in red giant stars, *Monthly Notices of the Royal Astronomical Society*, Volume 530, Issue 1, May 2024, Pages 149–166, doi:10.1093/mnras/stae820

Selected Presentations

Jan 2026 **Fast IMCOM: Unlocking New Scientific Frontiers with High-Fidelity Image Coaddition (contributed oral presentation)**, *247th Meeting of the American Astronomical Society*, Phoenix, United States.

Oct 2025 **Testing Stellar Theory with Joint Spectroscopic and Asteroseismic Catalogs (invited seminar talk)**, *AstroCoffee (University of Pittsburgh)*, Pittsburgh, United States.

Oct 2025 **Forecasts for weak lensing cosmology with Roman Space Telescope (contributed parallel talk)**, *28th International Conference on Particle Physics and Cosmology (COSMO-25)*, Pittsburgh, United States.

Aug 2025 **PyIMCOM Hack Session 2025**, *Shear and Clustering Measurement Working Group Hack Week (Roman HLIS Cosmology PIT)*, Columbus, United States.

Jul 2024 **IMCOM: optimized image coaddition pipeline for Roman weak lensing cosmology (contributed plenary talk)**, *Challenging Theory with Roman: From Planet Formation to Cosmology*, Pasadena, United States. [Abstract (please search for "IMCOM"), Slides, Video]

Sep 2022 **Theoretical predictions concerning the first dredge-up**, *Summer Research Poster Session (The Ohio State University)*, Columbus, United States.

Aug 2019 **How to define a relaxed cluster?**, *Summer MKI Undergraduate Research Forum (SMURF; Massachusetts Institute of Technology)*, Cambridge, United States.

Teaching Experience

2022 – 2023 **Instructor, Online Python Tutorial**.
About 46 students in 4 cohorts. Gave lectures, designed exercises, and mentored personal projects.

2021 – 2023 **Teaching Associate (Grader), Physics 7501 & 7502: (Graduate) Quantum Mechanics**, *The Ohio State University*, Columbus, United States.

Miscellany Taught or tutored college mathematics (2017, 2018, 2020, 2023), college physics (2023), graduate school application (2022, 2023, 2024), French phonology (2017, 2018), etc.

References

Available on request.